Artificial Intelligent - Report Lab 1

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1 Idea

To ensure the vacuum agent can discover and clean all dirt cells, it must continuously explore and update its internal map of the environment. That is for each iteration, the agent identifies the nearest frontier cell that is currently `UNKNOWN` and sets it as the target destination. Then, a path-finding algorithm is applied to navigate the agent towards this goal, progressively expanding its knowledge of the world. During this process, whenever the vacuum agent detects a dirt cell, the agent will clean it immediately. Note that this method can be applied for both task 1 and task 2 requirements.

Here is the pseudo-code for the process:

```
if see dirt:
    do clean

frontier <- find_frontier() # BFS to find the next cell to visit

if can find frontier:
    # Expand the internal map
    path <- a_star(current_position, frontier)
    do follow_path(path)
else if not at home:
    # Since we already found and visited all possible frontier -> 100% there is nothing to clean -> return home
    path <- a_star(current_position, home)
    do follow_path(path)
else:
    # Already at home
    do nothing
```

You can see the Python implementation of this pseudo-code at lines 190-224 of `myvacuumagent.py`, also the `a_star()` (251-279) and `find_frontier()` (310-335) function.

2 Performance Evaluation

My Vacuum Performance

With the `iteration_counter = height * width * 2`, My Vacuum agent gives the performance score at 650.0, and successfully returned to home after cleaning all the dirt. However, this configuration does not guarantee that the agent will always successfully return home after completing the cleaning process.

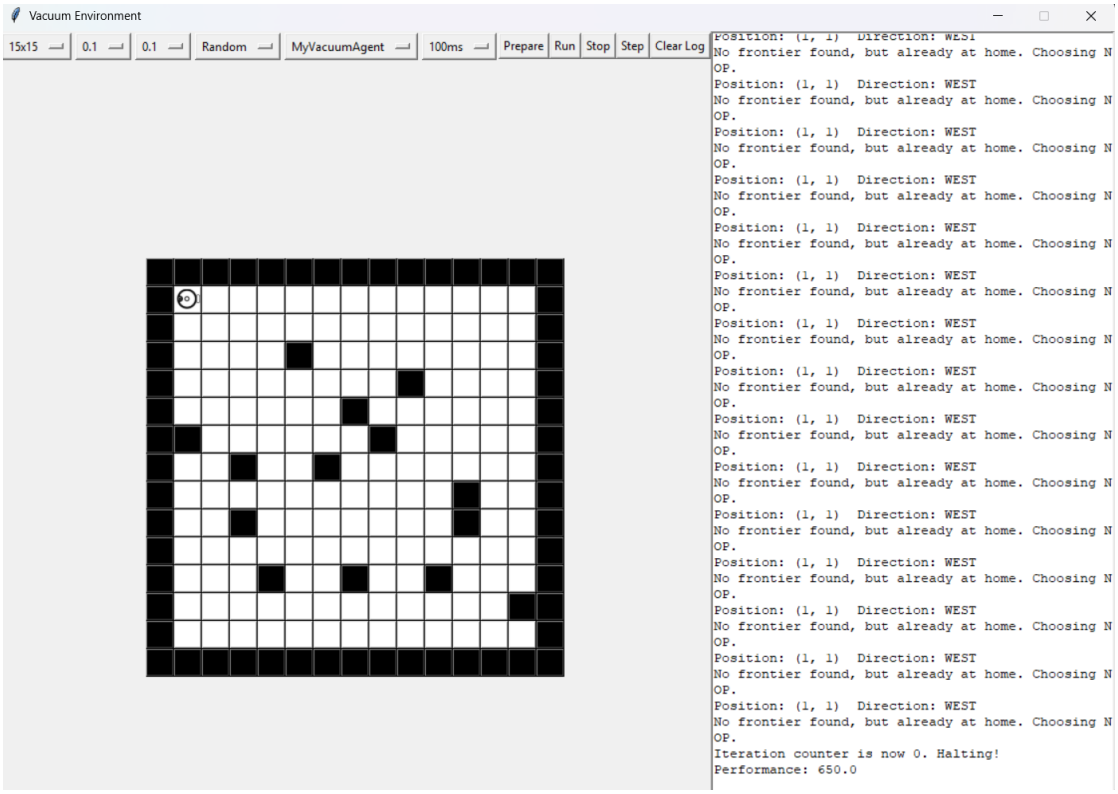

Figure 1: My Vacuum Performance

Reactive Agent Performance

Compared to the My Vacuum agent, the Reactive agent produces a significantly more modest result, achieving a performance score of only 55.0 and failing to clean all the dirt in the environment.

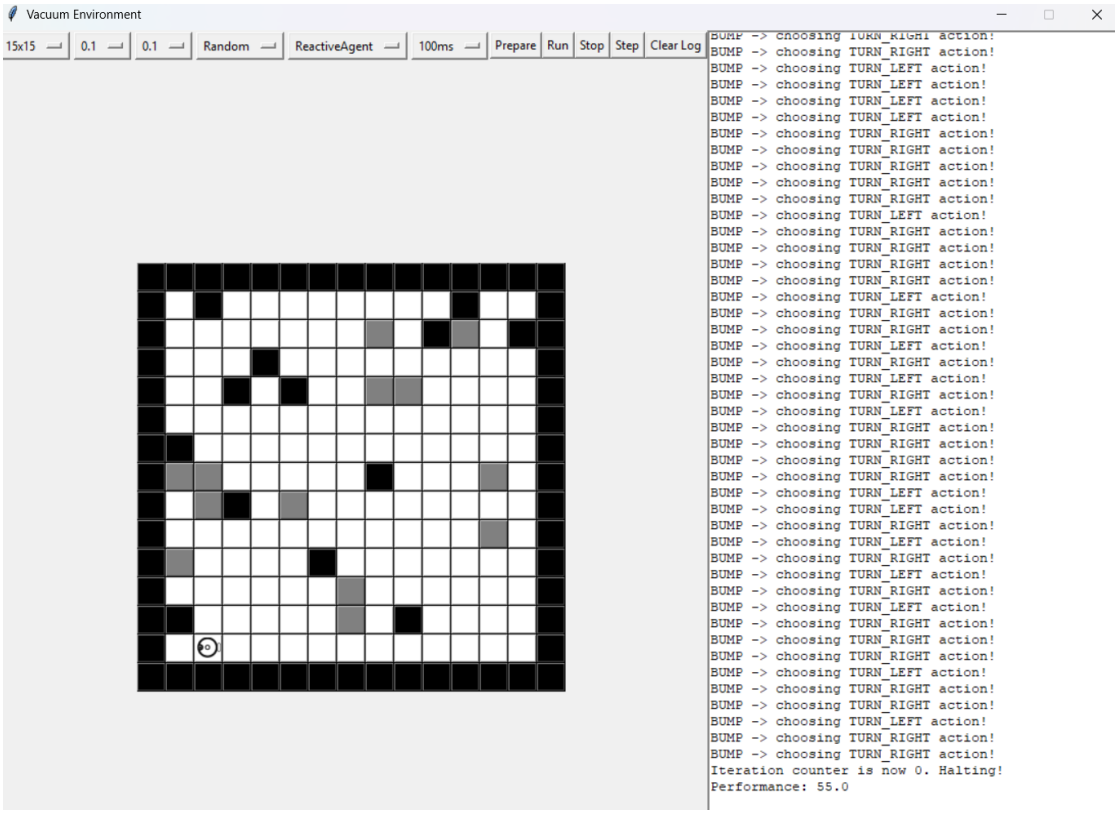

Figure 2: Reactive Agent Performance